

Circular coil

Verification of Biot Savart's law and determination of B_h

Aim:-

1. To study the variation of the magnetic field along the axis of a circular coil carrying current as predicted by Biot Savart's law
2. To determine the value of the horizontal component of the earth's magnetic field B_h

Apparatus:

Circular coil apparatus, an ammeter, a commutator, a rheostat and a battery (2V). The circular coil apparatus consists of a few turns of insulated copper wire wound over a wooden circular frame which is fixed vertically. A horizontal wooden base passes through the circular frame. A compass box is placed on the platform with its centre along the axis of the coil. This can be moved along the length of the horizontal wooden base and can be fixed at any position.

Principle:

When a current passes through a coil, a magnetic field is produced along the axis of the coil. If the magnetic flux density on the axis of the circular coil B and the earth's horizontal flux density B_H , act on a magnetic needle in perpendicular directions, then we have

$$B = B_H \tan \theta, \text{ called tangent law.}$$

The flux density B at a point on the axis of a circular coil is given by

$$B = \frac{\mu_0 N I a^2}{2(a^2 + x^2)^{\frac{3}{2}}}$$

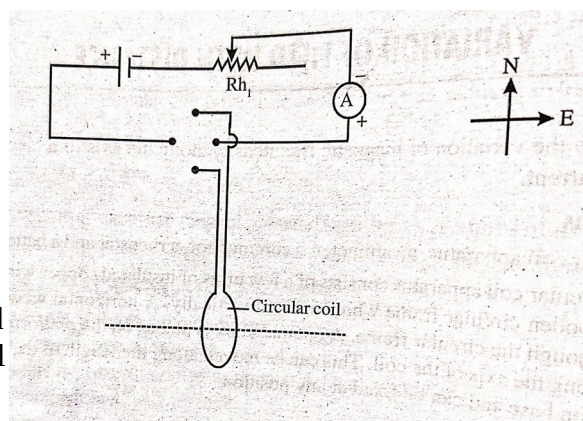
where N = number of turns of the coil, I = current through the coil

a = the radius of the coil, x = the distance of the point from the centre of the coil

This expression shows that when x varies B also varies. When B varies $\tan \theta$ varies, since $B \propto \tan \theta$. Thus to study the variation of B with distance we measure $\tan \theta$ for various distances. A graph drawn between distance x and $\tan \theta$ tell us how does B vary with distance.

Procedure:

Connections are made as shown in figure. To make B and B_H perpendicular, following adjustments must be made before starting the experiments. The compass box is kept at the centre of the circular coil ($x = 0$). The plane of the coil is adjusted to be along the magnetic meridian. For this the plane of the coil is kept roughly in the north-south direction. Rotate the compass box alone till 90 - 90 line becomes parallel to the plane of the coil, then the whole apparatus is rotated till the aluminium pointer reads 0-0. Now the horizontal wooden base is exactly in the east west directions.



After having made the preliminary adjustments, we can say the circular coil is ready for use. The circuit is switched on and the rheostat is adjusted so that the deflection is about 60° . The

readings of the both ends of the aluminium pointer are taken. The current is reversed and the readings are taken again. The average of the four readings gives θ corresponding to the distance $x = 0$.

The compass box is placed on one side of the coil along the axis at a distance x (say 3cm) from the centre of the coil on the eastern side of the coil with the help of a wooden stand. The circuit is closed and the deflections of both ends of the aluminium pointer are taken. The current through the coil is reversed and two more readings are taken. The average is taken and the compass box is transferred to the west side of the coil, four more deflections are taken. The average of four deflections gives θ corresponding to $x = 3\text{cm}$. Repeat the experiment by increasing the distance x . We can continue this till the deflections may go upto 30° . Calculate $\tan\theta$ corresponding to each distance for each side separately. A graph is plotted between distance x and $\tan\theta$. This tells us how does the field (B) vary with distance

Observations:

Current in the Ammeter, $I = \dots\dots\dots\text{A}$

Radius of the coil, $a = \dots\dots\dots\text{m}$

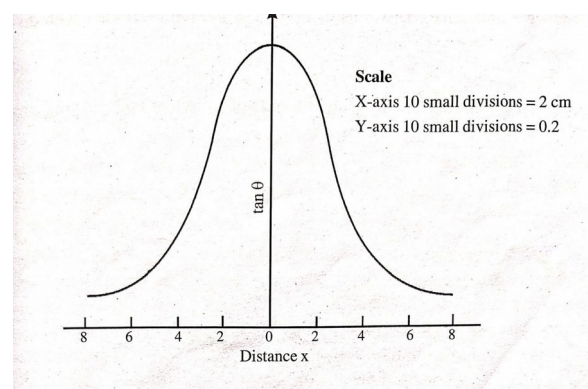
Number of turns, $N = \dots\dots\dots$

Distance x in cm	On the Eastern side				On the Western side				Mean θ	$\tan\theta$	$\frac{I}{\tan \theta}$
	Deflections				Deflections						
	θ_1	θ_2	θ_3	θ_4	θ_1	θ_2	θ_3	θ_4			

Mean $\frac{I}{\tan\theta} = \dots\dots\dots$

$$B_H = \frac{\mu_0 N a^2}{2(a^2 + x^2)^{\frac{3}{2}}} \left(\frac{I}{\tan\theta} \right) = \dots\dots\dots$$

$$\mu_0 = 4\pi \times 10^{-7}$$



Result:

1. Variation of magnetic flux density along the axis of a circular coil carrying current has been studied.
2. The value of the horizontal component of the earth's magnetic field $B_H = \dots\dots\dots\text{T}$